

Faizan Mohammed Jamal Shaikh

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PROFESSIONAL SUMMARY

Computer Science Honors student at NJIT (Dean's Scholar, Albert Dorman Honors College) with hands-on experience across AI/ML, information retrieval, and distributed infrastructure. Comfortable working as a versatile, full-stack engineer, from architecting agentic RAG pipelines and fine-tuning language models to building and hardening production web services on Docker/Kubernetes. Practicing DSA daily, automates routine work with Python, and an open source contributor in my free time.

EDUCATION

New Jersey Institute of Technology — BS in Computer Science with specialization in AI

Expected May 2029

Dean's Scholar, Albert Dorman Honors College | GPA: 3.85/4.00

Newark, NJ

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Data Science, Discrete Math, Linear Algebra

EXPERIENCE

Undergraduate Research Assistant

January 2026 – Present

New Jersey Institute of Technology

Newark, NJ

- Build a Clinical Ontology Evaluation Framework in Python, Pandas, and REST APIs to benchmark LLMs on 400+ SNOMED CT concepts.
- Design and automate LLM-as-ontology-server evaluation pipelines with a PhD researcher, using Git to version-control 20+ experiment iterations.

IT Support Specialist (Tier 2)

March 2026 – Present

New Jersey Institute of Technology

Newark, NJ

- Provide Tier 2 technical support and hardware management for university infrastructure, resolving 100+ escalated tickets.
- Diagnose and document root-cause fixes for escalated software, hardware, and network issues, partnering with Tier 1 staff to reduce repeat incidents.

Data Analyst and Office Assistant

March 2026 – Present

NJIT, Office of the President

Newark, NJ

- Performed Data Analytics and managed confidential records to support executive meetings and institutional decision-making, preparing recurring reports for university leadership.

PROJECTS

Agentic RAG Pipeline (ARAG)

May 2026 – Present

LangGraph, Supabase (pgvector), sentence-transformers(BAAI), NVIDIA NIM, Python

- Architected a 6-node agentic RAG system in LangGraph (retrieve, grade, rewrite, generate, verify, fallback) that grades and verifies grounding before answering, cutting hallucinations versus naive retrieve-then-generate RAG.
- Built a retrieval layer on Supabase pgvector with a 384-dimensional HNSW index and local sentence-transformer embeddings, returning the top-6 ranked chunks per query with inline citations via the NVIDIA NIM API.
- Implemented automatic query rewriting (up to 3 retries) and an LLM fallback path, keeping the system able to answer even when the document corpus lacks direct coverage.

Small-Language-Model (SLM) Fine-Tuning Pipeline

June 2026 – Present

Designing a specialized "expert model" by fine-tuning a compact 3B parameter/weights model Llama-3B on curated open-source repositories and licensing metadata to auto-architect zero cost product tech stacks, matching 70B model recommendation quality. Implemented QLoRA & DPO Alignment via Unsloth and Hugging Face TRL to minimize architectural hallucination and enforce free/permissive-tier constraint compliance.

Synthesized Synthetic Instruction Pairs mapping high-level software requirements to optimal open-source tools, framework tradeoffs, and deployment paths within hardware budget limits.

Xyris Vision — xyrisvision.vercel.app

June 2026

React, Supabase, Stripe, Computer Vision

- Built a privacy-first image redaction web app in 24 hours at the NJx Hackathon, acquiring 50+ active users.
- Engineered a client-side pipeline to scrub EXIF metadata and redact PII using computer vision, ensuring zero server-side data persistence for 100+ images processed.
- Integrated Stripe for usage-based billing and Supabase for secure backend authentication.

Collab IDE

June 2026

Next.js, React, Python, Docker, Electron

- Engineered a desktop IDE application by forking VS Code with Electron and Next.js, enabling multi-developer real-time collaboration on unified software projects.
- Containerized workspace environments using Docker, ensuring synchronized package dependencies across all collaborators. Integrated a multi-agent AI orchestration engine that pools individuals AI agents to optimize context sharing and reduce overall LLM token usage.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, SQL, HTML/CSS, PHP

AI/ML: LLMs, RAG, Agentic AI, Information Retrieval, NLP, Model Context Protocol (MCP), Fine-Tuning (QLoRA, DPO), PyTorch, TensorFlow, Generative AI

Systems & Infrastructure: Docker, Kubernetes, Distributed Systems, Linux/Unix, Git, REST APIs, PostgreSQL, Supabase, GitHub

Frameworks & Tools: React, Node.js, Next.js, Flask, Django, TailwindCSS, Pandas, NumPy, Vercel, Stripe, Render